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Data Science Capstone
Capstone Topic Proposal
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Initial Overview

With our ever-changing world–especially in the realms of artificial intelligence–the need for growing data centers and technologies has never been understated. While a few select Western companies have been driving this initiative, the supposedly most important players are those providing the resources. Intermediate mineral suppliers, the "middlemen," as some might say, have played a crucial part in global trade. From the Silk Road to the AI Bubble, they have been the ones "supplying the shovels" to various companies, governments, and industries.

SKY, Samsung, Micron, SK Hynix, etc., have been producing DRAM, silicon chips, and other crucial components important for all of today's electronics; however, AI companies have been taking the majority of the world's supply of RAM. This has impacted the overall price for consumers and even larger companies in other sectors looking to purchase DRAM and other related parts at a standstill. NVIDIA–one of the pioneers of AI and the largest graphics and video card producer–revealed that for the first time in nearly two decades, it will not release any new graphics cards due to the DRAM shortage (Bloomberg 2025). Like OPEC or other cartels, there have been debates about creating a centralized organization of the leading chipmakers to curb the volatility in the sector (Bloomberg 2019).

Data Analytic Method(s):

The data I plan to analyze for my capstone will come straight from the Federal Reserve Bank of St. Louis, as they have already established the Semiconductor Producer Price Index (PPI), which illustrates the rise of semiconductor prices throughout the "AI Boom". I will also be analyzing the International Energy Agency (IEA)'s *Global Data-center Electricity Demand* dataset to reveal how data-center electricity usage correlates with the growing demand for semiconductor chips. Additionally, but to a much lower extent, I will be looking at interest rates from the Bureau of Labor Statistics/Federal Reserve as well.

To be frank, my methodology is still a work in progress, but here is what I am thinking: I will combine monthly semiconductor price data with data-center electricity demand and interest rate data from 201 -2024 (2014 was when the IEA started collecting global data-center electricity

demand). I will then use a series of regression models that hold and, potentially, weigh different variables to see significance. Here are some examples of the models in R I might be using; these examples come from a prior paper that analyzed a similar topic:

Model 1: Logged Data-Center Demand Alone

```
model_1_alone <- lm(
  log_ppi ~ log_data_center_twh,
  data = monthly_data
)

summary(model_1_alone)
```

Model 2: Add Interest Rate Control

```
model_2_interest <- lm(
  log_ppi ~ log_data_center_twh + interest_rate,
  data = monthly_data
)

summary(model_2_interest)
```

Model 3: Add Post-2021 Indicator

```
model_3_post <- lm(
  log_ppi ~ log_data_center_twh + interest_rate + post_2021,
  data = monthly_data
)

summary(model_3_post)
```

Model 4: Interaction Model

```
model_4_interaction <- lm(
  log_ppi ~ log_data_center_twh * post_2021 + interest_rate,
  data = monthly_data
)

summary(model_4_interaction)
```

While these are still my elementary models and will undoubtedly change throughout the course, I hope these examples will serve as the "backbone" to my model and analysis. If you have any questions or suggestions, please let me know!

Current Sources & Data Sets:

"AI Boom Fuels DRAM Shortage and Price Surge." *IEEE Spectrum*, 2026.

Federal Reserve Bank of St. Louis. (2026). *Producer Price Index by Industry: Semiconductor and other electronic component manufacturing*. FRED.

Glencross, A. (2024). The geopolitics of supply chains: EU efforts to ensure security of supply. *Global Policy, 15*(4), 729–739. https://doi.org/10.1111/1758-5899.13388

International Energy Agency. (2026). *Data centre electricity use surged in 2025, even with tightening bottlenecks driving a scramble for solutions*.

Jin, H., & Yang, H. (2026, April 30). Samsung chip profit jumps almost 50-fold; supply shortage to worsen in 2027. *Reuters*.

Organisation for Economic Co-operation and Development. (2025). *Mapping the semiconductor value chain*.

Yang, H., & Jin, H. (2026, January 29). Apple warns memory costs are starting to bite as Samsung, SK Hynix prioritise AI chips. *Reuters*.